

Johannes Gaus

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Research and Professional Experience

- June 2025 – Present **Ph.D. Student & IMPRS-IS Scholar**, *NeuRoMech Lab (Neuromechanics and Rehabilitation Robotics Group), Hertie Institute for Clinical Brain Research and Center for Integrative Neuroscience, University of Tübingen*
Research on uncertainty in machine learning and intention prediction for reliable, human-centered assistive robotics.
- Sept. 2024 – May 2025 **Working Student Research**, *Deutsches Zentrum für Luft- und Raumfahrt (DLR)*
Master's Thesis: Studying the Explainability of Uncertainties in Point Cloud Registration for Robotic Systems.
- Jan. 2025 – May 2025 **Research Assistant**, *KIT Institut für Angewandte Informatik und Formale Beschreibungsverfahren (AIFB)*
Contributed to research projects in software engineering and machine learning (Python & GNN). Implemented data preprocessing pipelines and ML training scripts.
- Apr. 2024 – Sept. 2024 **Working Student Data Science**, *Medicalvalues GmbH*
Participated in agile software development for laboratory data analysis. Designed and prototyped machine learning models for predictive analytics in medical diagnostics.
- Aug. 2023 – Mar. 2024 **Working Student Software Development**, *TeamViewer*
Supported the product owner in software projects. Conducted quality assurance testing and assisted the data science team with exploratory analysis and feature engineering.
- Feb. 2023 – Jul. 2023 **Working Student Software Development**, *KubeOps*
Agile Software Development in C# and Go in the Kubernetes Domain.
- May 2021 – Jan. 2023 **Working Student Software Development and Data Science**, *Campus Schwarzwald*
Contributed to a range of research projects, including software development, internal IT, IT security, and machine learning.

Research and Professional Experience (continued)

- May 2020 – May 2022  **Student Assistant**, *University of Tübingen, Institute of Education Sciences*
- Sept. 2020 – Nov. 2020  **Student Assistant**, *Zentrum für Schulqualität und Lehrerbildung Baden-Württemberg (ZSL BW)*
- Oct. 2019 – Jul. 2020  **Mathematics Tutor**, *Studienkreis GmbH*

Education

- Feb. 2026 – Present  **IMPRS-IS Scholar**
International Max Planck Research School for Intelligent Systems (IMPRS-IS), University of Tübingen and Max Planck Institute for Intelligent Systems.
- June 2025 – Present  **Ph.D. in Computer Science, University of Tübingen**
Neuromechanics & Rehabilitation Robotics Group (NeuRoMech) at the Hertie Institute for Clinical Brain Research & Center for Integrative Neuroscience, University of Tübingen
- Oct. 2022 – May 2025  **M.Sc. in Computer Science, Karlsruhe Institute of Technology (KIT)**
Specialization: Software Engineering & Machine Learning.
Thesis title: *Studying the Explainability of Uncertainties in Point Cloud Registration for Robotic Systems.*
- Mar. 2019 – Aug. 2022  **B.Sc. in Computer Science, University of Tübingen**
Minor: Psychology.
Thesis title: *Probabilistic Short-term Wind Power Forecasting.*
- Oct. 2017 – Nov. 2018  **Business Informatics, Duale Hochschule Baden-Württemberg**

Skills

- Languages  German (native), English (C2), French (B1).
- Programming  Python, Java, C, C++, R, SQL, PostgreSQL, PHP, Matlab, VBA, HTML/CSS, ABAP.
- Tools & Technologies  Linux, Windows, Git, Docker, Kubernetes, ROS, Flask, .NET, Android, PyTorch, JIRA, Confluence, UPPAAL, Looker, LimeSurvey, LaTeX, SLURM, Active Directory.

Teaching, Service, and Outreach

- 2025–Present  **Tutor, Mathematics for Neuroscience** – University of Tübingen. Conducted tutorial sessions, explained mathematical foundations for neuroscience students, and supported exam preparation.

Teaching, Service, and Outreach (continued)

- **Instructor and Project Lead, Software Project for Bachelor Students** – University of Tübingen. Designed, organized, and led a lecture-based software project focused on developing tremor detection software for clinical applications; supervised student teams and guided full project lifecycles.
- **Master's Thesis Advisor** – University of Tübingen. Advised multiple Master's students on research projects, methodology, and scientific writing.

Awards and Achievements

- 2024 ■ **Summer School: Modeling Infectious Diseases and Health Economics** – University of Antwerp.
- 2015 ■ **1st Place, Jugend Hackt Ulm** – “Mit Code die Welt verbessern.”

Extracurricular Activities

- 2021–Present ■ **Volunteer MINT Botschafter, MINT Netzwerk Deutschland**, Manage social media channels and participate in the committee evaluating schools' applications for MINT programs.
- 2022–2025 ■ **Team Lead, Hochschulgruppe KITcar, Navigation Subteam**. Led the navigation subteam developing autonomous navigation modules using ROS and C++; participated in academic competitions.

Research Publications

Publications and Preprints

- 1 J. A. Gaus, J. P. F. Charaja, and D. Haeufle, *Confidence-gated robot autonomy: When does uncertainty actually help?*, ICRA Workshop paper, 2026. arXiv: 2605.18045 [cs.R0].
- 2 J. A. Gaus, D. Häufle, and W.-J. Baek, *Super – a framework for sensitivity-based uncertainty-aware performance and risk assessment in visual inertial odometry*, ICRA Workshop paper, 2026. arXiv: 2512.14189 [cs.R0].
- 3 J. A. Gaus, W. Ilg, and D. Haeufle, *When to act: Calibrated confidence for reliable human intention prediction in assistive robotics*, 2026. arXiv: 2601.04982 [cs.R0].
- 4 J. A. Gaus, L. Schneider, Y. Shi, J. Lee, R. Rayyes, and R. Triebel, *Human-interpretable uncertainty explanations for point cloud registration (ICP, GP-CA)*, Accepted at ICRA 2026, Vienna, 2026. arXiv: 2509.18786 [cs.R0].
- 5 J. A. Gaus, J. Yoon, W.-J. Baek, S. Choi, S. Park, and J. Park, *Guard: Toward a compromise between traditional control and learning for safe robot systems*, Under review at IROS, 2026. arXiv: 2509.23312 [cs.R0].